

ChainSecurity
Security Audit Report: Sample DEX

[CS-1] Missing access control on admin setter

Critical

Description

The setFeeRecipient function in contracts/DEX.sol:42 has no onlyOwner guard. Any caller can redirect the protocol fee.

Recommendation

Add an onlyOwner modifier to setFeeRecipient.

Acknowledgement

Code Corrected

[CS-2] Integer overflow in fee calculation

High

Description

The fee calc in contracts/DEX.sol:200 multiplies unchecked uints.

Recommendation

Use Solidity 0.8+ checked arithmetic.

Acknowledgement

Code Corrected

[CS-3] Incorrect event emission order

Medium

Description

Events in contracts/DEX.sol:300 emit before state is updated.

Recommendation

Move the emit to after the state update.

Acknowledgement

Acknowledged

[CS-4] Missing zero-address check on token setter

Low

Description

The setToken function in contracts/DEX.sol:400 lacks a zero check.

Recommendation

Add `require(newToken != address(0))` guard.

Acknowledgement

Risk Accepted

The team prefers to keep the function unconstrained for upgrade flexibility.

[CS-5] Inconsistent comment style across modules

Informational

Description

Comments in `contracts/DEX.sol` mix NatSpec and inline conventions.

Recommendation

Standardise on NatSpec for all public/external functions.

Acknowledgement

Acknowledged